

1. K-Nearest Neighbors (KNN)

1. A KNN model compares two customers using age and weekly usage. Which equation gives the Euclidean distance between customers $A = (a_1, u_1)$ and $B = (a_2, u_2)$?

- A. $d(A,B) = \sqrt{[(a_1 - a_2)^2 + (u_1 - u_2)^2]}$
- B. $d(A,B) = (a_1 + a_2) / (u_1 + u_2)$
- C. $d(A,B) = a_1u_1 + a_2u_2$
- D. $d(A,B) = |a_1 - a_2| \times |u_1 - u_2|$

2. A new customer's distances to three training customers are 2, 5, and 8. Their churn labels are 1, 0, and 0, respectively. For $K = 1$, what does the KNN voting equation use?

- A. The label of the customer at distance 2
- B. The majority label among all three customers
- C. The average of the three distances
- D. The label of the customer at distance 8

3. For KNN regression with $K = 3$ and uniform weights, the three nearest houses have prices of ₹40 lakh, ₹50 lakh, and ₹60 lakh. Which expression gives the predicted price?

- A. $(40 + 50 + 60) / 3$ lakh
- B. $(40 \times 50 \times 60) / 3$ lakh
- C. $40 + 50 + 60$ lakh
- D. $\max(40, 50, 60) - \min(40, 50, 60)$ lakh

4. A KNN model uses Manhattan distance for two features. Customer A has values (4, 9), and customer B has values (7, 5). Which expression calculates their distance?

- A. $|4 - 7| + |9 - 5|$
- B. $\sqrt{[(4 - 7)^2 + (9 - 5)^2]}$
- C. $(4 + 7) \times (9 + 5)$
- D. $(4 - 7) \times (9 - 5)$

5. A dataset contains customer age ranging from 18 to 70 and monthly fee ranging from ₹200 to ₹1,500. KNN uses a distance equation containing both features. Why might scaling be necessary?

- A. The larger numerical range of monthly fee could dominate the distance
- B. Scaling changes the churn labels from 0 and 1 to customer IDs
- C. Euclidean distance cannot be calculated with two features
- D. Scaling makes every customer's distance equal

2. Naive Bayes

1. A model classifies an email as spam (S) using its words (W). Which expression represents the probability the model wants to estimate?

- A. $P(S | W)$
- B. $P(W | S)$ only
- C. $P(S) \times P(W)$ only
- D. $P(S + W)$

2. Bayes' theorem can be written as $P(S | W) = P(W | S)P(S) / P(W)$. In this equation, what does $P(S)$ represent?

- A. The prior probability of spam
- B. The probability of the words given spam
- C. The distance to the nearest spam email
- D. The model's classification accuracy

3. In a training set of 200 emails, 50 are spam. If the model estimates the spam prior from this dataset, which expression does it use?

- A. $P(\text{spam}) = 50 / 200$
- B. $P(\text{spam}) = 200 / 50$
- C. $P(\text{spam}) = 50 \times 200$
- D. $P(\text{spam}) = 200 - 50$

4. A simplified Naive Bayes calculation uses $P(x_1, x_2 | \text{class}) \approx P(x_1 | \text{class}) \times P(x_2 | \text{class})$. What assumption allows this multiplication?

- A. The features are conditionally independent given the class
- B. The two features always have identical values
- C. Both classes have equal numbers of samples
- D. The features are continuous rather than categorical

5. A word never appeared in spam emails during training. The model uses Laplace smoothing: $P(\text{word} | \text{spam}) = (\text{count}(\text{word in spam}) + \alpha) / (\text{total spam word count} + \alpha V)$, where V is vocabulary size. Why is α added?

- A. To avoid assigning zero probability solely because the word was unseen
- B. To increase the number of spam emails in the dataset
- C. To calculate Euclidean distance
- D. To set the classification threshold to α

3. Decision Trees

1. A tree node contains 8 customers who stayed and 2 who canceled. Its Gini impurity is calculated as $Gini = 1 - (8/10)^2 - (2/10)^2$. What does this equation measure?

- A. How mixed the two classes are at the node
- B. The distance between the ten customers
- C. The number of branches in the whole tree
- D. The probability that every customer will cancel

2. A decision tree considers splitting a node using $weekly_usage_hours \leq 3$. What is the purpose of comparing impurity before and after the split?

- A. To assess whether the split creates more useful class separation
- B. To calculate the average age of all customers
- C. To select the nearest customer
- D. To determine the number of trees in a forest

3. Entropy for two classes can be written as $H = -p_0 \log_2(p_0) - p_1 \log_2(p_1)$. If a leaf contains only customers who stayed, which values describe its class proportions?

- A. $p_0 = 1$ and $p_1 = 0$
- B. $p_0 = 0.5$ and $p_1 = 0.5$
- C. $p_0 = 1$ and $p_1 = 1$
- D. $p_0 = 0$ and $p_1 = 0$

4. A split produces two child nodes containing 30 and 10 customers. Which expression calculates their weighted average impurity?

- A. $(30/40) \times \text{impurity}(\text{left}) + (10/40) \times \text{impurity}(\text{right})$
- B. $\text{impurity}(\text{left}) + \text{impurity}(\text{right}) + 40$
- C. $30 \times 10 \times \text{impurity}(\text{left})$
- D. $[\text{impurity}(\text{left}) - \text{impurity}(\text{right})] / 40$

5. A customer reaches a tree leaf containing 12 customers who canceled and 8 who stayed. Which expression could be used as that leaf's estimated churn probability?

- A. $12 / (12 + 8)$
- B. $8 / 12$
- C. $(12 + 8) / 12$
- D. 12×8

4. Random Forest

1. Five trees predict churn labels $[1, 1, 0, 1, 0]$ for a customer. A random forest uses majority voting. Which expression describes the final class?

- A. $\text{mode}(1, 1, 0, 1, 0)$
- B. $\max(1, 1, 0, 1, 0) - 1$
- C. $1 \times 1 \times 0 \times 1 \times 0$
- D. The label predicted by the first tree only

2. Four trees in a random forest regressor predict customer spending of ₹800, ₹1,000, ₹900, and ₹1,100. Which expression gives the forest's usual regression prediction?

- A. $(800 + 1,000 + 900 + 1,100) / 4$
- B. $800 \times 1,000 \times 900 \times 1,100$
- C. $\max(800, 1,000, 900, 1,100)$
- D. $1,100 - 800$

3. A forest contains 100 trees. For a particular customer, 72 trees vote for churn. Which simple fraction represents the proportion of trees voting for churn?

- A. $72 / 100$
- B. $100 / 72$
- C. $100 - 72$
- D. 72×100

4. A forest is trained using bootstrap samples. Which equation-like description matches sampling *with replacement*?

- A. A training customer can appear more than once in one tree's sample
- B. Each training customer must appear exactly once in every tree's sample
- C. Every tree trains only on the test set
- D. Training customers are selected according to their predicted labels

5. A tree's out-of-bag prediction is evaluated using training customers excluded from *that tree's* bootstrap sample. Which expression identifies an out-of-bag customer for that tree?

- A. Customer $\in$ training data AND customer $\notin$ that tree's bootstrap sample
- B. Customer $\in$ final test set only
- C. Customer $\in$ that tree's bootstrap sample twice
- D. Customer has churn label 1

5. Logistic Regression

1. A logistic regression model computes $z = b_0 + b_1x_1 + b_2x_2$ and then $p = 1 / (1 + e^{(-z)})$. What does p represent in a binary churn model?

- A. The estimated probability of the positive class, such as churn
- B. The number of trees required
- C. The distance to the nearest customer
- D. The number of input features

2. A model uses a threshold of 0.5 and estimates $P(\text{churn} = 1 \mid X) = 0.73$. Which rule assigns its class label?

- A. Predict churn if $p \geq 0.5$
- B. Predict churn only if $p = 1$
- C. Predict churn if $p < 0.5$
- D. Predict churn only if every feature is positive

3. The model's linear score is $z = -2 + 0.5 \times \text{support_tickets}$. For a customer with 4 support tickets, which expression gives z ?

- A. $-2 + 0.5 \times 4$
- B. $-2 / (0.5 \times 4)$
- C. $-2 \times 0.5 \times 4$
- D. $4 / (-2 + 0.5)$

4. The odds of churn are defined as $p / (1 - p)$. If $p = 0.75$, which expression gives the odds?

- A. $0.75 / 0.25$
- B. $0.25 / 0.75$
- C. $0.75 + 0.25$
- D. 0.75×0.25

5. Logistic regression can be written as $\log[p / (1 - p)] = b_0 + b_1x_1 + \dots + b_nx_n$. A fitted coefficient b_1 is positive. Holding the other features fixed, what does increasing x_1 do?

- A. It increases the predicted log-odds of the positive class
- B. It always decreases the predicted log-odds
- C. It increases the number of training samples
- D. It changes the model into a decision tree